# Gagana M D

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Early-career AI safety researcher working on monitoring and evaluating AI agents. I run controlled, preregistered experiments, build the tested code behind them, and work in small teams across time zones.

## Research Experience

**PRISM AI Safety Research Fellowship** – *Research Fellow, Remote (mentor: Jobst Heitzig)* May 2026 – Oct 2026

- Evaluated 410 LLM-generated world models: 134/135 games ran, yet structural F1 was only 0.366 (NeurIPS 2026 NEmo).
- Main code contributor: 78 of ∼116 commits to the shared codebase (experiment runners, graph-analysis library, tests).
- Supported two other fellowship teams (interpretability, AI control) with experiment tracking, reviews and milestones.

**Sentient Futures** – *Research Fellow, Remote (mentor: Julia Bossmann)* Sep 2026 – Present

- Designed and hosted a 90-min roundtable on post-AGI cooperation for ∼10 governance, safety and foresight experts.
- Building behavioural evaluations of self-directed curiosity in LLMs that hold across prompts and controls.

**BaseThesis Labs** – *AI Safety Researcher, Evaluations, Bengaluru* May 2026 – Jul 2026

- Built an offline agent benchmark for CFO and due-diligence tasks with 50 CI tests and a leakage guard. Code
- Compared 3 agent set-ups on 12 tasks: orchestration cost 3.5× latency and paid off only for coverage. Code

**Indian Institute of Science, DREAM Lab** – *Research Intern, Bengaluru* Jan 2025 – Aug 2025

- Co-authored a drone–fog co-scheduling paper: the heuristic gave 31–230% higher utility and 100% task completion.

**Neuromatch** – *Impact Scholar, Remote* Oct 2024 – Mar 2025

- Designed and ran a Matching Law study of transformer RL agents across reward ratios from 1:1 to 16:1.

## Selected Projects

**Detection Is Not Containment** – first author, Apart Research AI Incident Response Sprint Sep 2026

- Built a preregistered monitoring experiment: detection beat the last safe intervention point by 1 step. Code
- Showed fragility: only 8 of 12 meaning-preserving renames kept the decision; a false alarm made a benign agent fail.

**Collusion among LLM agents** – two team reports, Apart Research AI Collusion Research Sprint Oct 2026

- Ran probes, patching and steering on Qwen2.5-7B; a 1,656-run preregistered test found no punishment of undercutting.
- Led a covert monitor-swap study: bonus grabs fell to 0/40 (36/40 in controls), and 0/15 episodes spotted the swap.

## Publications and Reports

*Evaluating LLM-Generated World Models for Neuro-Symbolic Reasoning.* NEmo: Neuro-Symbolic Embodied Intelligence workshop, NeurIPS 2026. Accepted.

Co-author, two Apart Research collusion-sprint reports (2026): *Tacit Collusion in Pricing Agents* (ran the mechanistic experiments) Report; *Noticing the Move, Missing the Player* Report

**Gagana M D**, Ajay Kumar K, Vijayashekhar S S, Jignesh S. Bhatt. *Hyperspectral BiorNet: A Complete Wavelet Framework for Unmixing.* Wavelet-domain unmixing with a custom wavelet activation. IEEE InGARSS 2025, pp. 15–19. DOI

**Gagana M D**, Kaustubh Gupta, María Inés Berdiñas. *Detection Is Not Containment: Timely Monitoring Determines Whether AI Agent Failures Can Be Prevented.* Apart Research, 2026. Report

S. Raj, A. Khanda, **Gagana M D**, Y. Simmhan, S. K. Das. *Mobility-Aware Cost-Efficient Co-Scheduling of UAV Routes and Analytics across Edge and Fog Computing.* Manuscript.

Y. Karakoc, E. Sharifi, **M. D. Gagana**, Z. Sarayloo. *Investigating the Matching Law in Transformer-Based Reinforcement Learning Agents.* Zenodo, 2025. DOI

## Education

**B.E., Artificial Intelligence and Machine Learning** – Acharya Institute of Technology, CGPA 8.38/10 2021 – 2025

Also: Neuromatch Academy, Computational Neuroscience (2025) and Deep Learning (2024). Courses: AI Risk Fundamentals, Professional tier (Lens Academy, 2026); Future of AI (BlueDot Impact, 2026); ML Summer School (Cohere Labs, 2026); Red Teaming LLM Applications (DeepLearning.AI, 2026).

## Collaboration, Honours and Skills

**Collaboration:** co-led a 3-person sprint team; hosted a cross-disciplinary roundtable; Shakti Mentorship programs team and workshop facilitator (Women in Big Data India, 2026–); WiDS Ambassador; helped run 11+ Women Who Code workshops reaching 350+ people.

**Honours:** 2nd place, AMD AI Sprint Hackathon (AMD, Meta, Hugging Face); 1st place, Federated Learning Client Selection Challenge, IISc.

**Tools:** Python, PyTorch, Transformers, Stable-Baselines3, Ollama, pytest, GitHub Actions, Docker, FastAPI, Git, Linux.